$\Rightarrow \alpha + 2\beta = a$
$\alpha + 3\beta = b$

$\Rightarrow \beta = b - a$
$\alpha = a - 2b + 2a$
$= 3a - 2b$

$\therefore T(w_1) = 4u_1 + 2u_2 = 8w_1 - 2w_2$
$T(w_2) = 9u_1 + 8u_2 = 11w_1 - w_2$

$\therefore B = [T]_{S'} = \begin{pmatrix} 8 & 11 \\ -2 & -1 \end{pmatrix}$

Now let's do the change of basis matrix from S to S'

$w_1 = u_1 + u_2$
$w_2 = 2u_1 + 3u_2$

$\therefore P = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$

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OR

$w_1 = u_1 + u_2$
$w_2 = 2u_1 + 3u_2$

$\Rightarrow 2w_1 = 2u_1 + 2u_2$
$w_2 = 2u_1 + 3u_2$

$u_2 = w_2 - 2w_1$

$u_1 = w_1 - w_2 + 2w_1 = 3w_1 - w_2$

$P^{-1}AP = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$
$= \begin{pmatrix} 13 & -5 \\ -5 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$
$= \begin{pmatrix} 8 & 11 \\ -2 & -1 \end{pmatrix} = B$

~~$T(w_1) = 3w_1 - 2w_2 \quad 3(3w_1 - w_2) - 2(w_2 - 2w_1) = 13w_1 - 5w_2$~~

$T(w_1) = 4u_1 + 2u_2 = 4(3w_1 - w_2) + 2(w_2 - 2w_1)$
$= 8w_1 - 2w_2$

$T(w_2) = 9u_1 + 8u_2 = 9(3w_1 - w_2) + 8(w_2 - 2w_1)$
$= 11w_1 - w_2$

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